expectancy of a child of an old mother was less than that of a child of a young mother. This means that, if a woman had a child and a grandchild born on the same day, the grandchild could expect to live longer than the child. When a woman reached the age where the average chance of each child reaching adulthood was just less than half the chance of each grandchild of the same age reaching adulthood, any gene for investing in grandchildren in preference to children would tend to prosper. Such a gene is carried by only one in four grandchildren, whereas the rival gene is carried by one in two children, but the greater expectation of life of the grandchildren outweighs this, and the 'grandchild altruism' gene prevails in the gene pool. A woman could not invest fully in her grandchildren if she went on having children of her own. Therefore genes for becoming reproductively infertile in middle age became more numerous, since they were carried in the bodies of grandchildren whose survival was assisted by grandmotherly altruism.

This is a possible explanation of the evolution of the menopause in females. The reason why the fertility of males tails off gradually rather than abruptly is probably that males do not invest so much as females in each individual child anyway. Provided he can sire children by young women, it will always pay even a very old man to invest in children rather than in grandchildren.

So far, in this chapter and in the last, we have seen everything from the parent's point of view, largely the mother's. We have asked whether parents can be expected to have favourites, and in general what is the best investment policy for a parent. But perhaps each child can influence how much his parents invest in him as against his brothers and sisters. Even if parents do not 'want' to show favouritism among their children, could it be that children grab favoured treatment for themselves? Would it pay them to do so? More strictly, would genes for selfish grabbing among children become more numerous in the gene pool than rival genes for accepting no more than one's fair share? This matter has been brilliantly analysed by Trivers, in a paper of 1974 called 'Parent-Offspring Conflict'.

A mother is equally related to all her children, born and to be born. On genetic grounds alone she should have no favourites, as we have seen. If she does show favouritism it should be based on differences in expectation of life, depending on age and other things. The mother, like any individual, is twice as closely 'related' to herself as she is to any of